



Validating the Corral Occupancy Sensor: A Manual Headcount Against the Count Setting Next Term's Bike-Rack Capacity

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Abstract. Facilities plans next term's bike-rack capacity expansion using each share-scooter corral's occupancy reading from an automated infrared sensor array installed across the University's corral network. We ask whether that reading tracks how many scooters a corral holds at a given moment or how long the scooters already there have stayed. Over nine weeks we set 15 of the University's 21 registered corrals against an unannounced manual headcount audit, timing 312 physical counts against the sensor's own smoothed occupancy index for the same minute. The two measures correlate only moderately (Spearman's $\rho = 0.49$; a fitted linear model explains 22% of headcount variance), and the shortfall appears systematic rather than noisy: corrals in the top tercile of mean scooter dwell time read, on average, 1.6 index points higher than headcount alone would predict, and corrals in the bottom tercile read 1.1 points lower. Of the five corrals the index ranks highest this term, and toward which Facilities' capacity review directs additional bay expansion, only two also rank in the top five by manual headcount. Removing the index's smoothing window and recomputing occupancy from instantaneous samples leaves the median per-corral correlation with headcount unchanged (0.35 smoothed, 0.35 unsmoothed), suggesting the shortfall sits in what the sensor observes rather than how it averages. We suggest the current index is read more reliably as a dwell-time indicator than an occupancy indicator, at least in the corrals and term we sampled.

Introduction

Slop University's share-scooter network relies on an infrared occupancy sensor installed along the kerb rail of each of the University's 21 registered corrals, sampling presence in every bay every ninety seconds and publishing a smoothed occupancy index to Facilities' capacity-planning dashboard. Each term, the index feeds directly into a single decision: which corrals receive additional bay capacity when the network's allocation is revised. No audit of the index against a physical count has been conducted since the sensor network's installation.

The gap is not a small one to leave open. An occupancy sensor built to answer "how many scooters are here" and a sensor built to answer "how long has something been here" can, in

principle, produce near-identical readings under steady, low-turnover conditions and diverge once turnover rises — a corral that cycles through many scooters briefly each hour looks, to a dwell-weighted instrument, emptier than a corral holding a handful of long-parked vehicles for days. Whether the University's sensor network sits closer to the first design or the second has, until this audit, been a question nobody with access to the capacity-planning process had put to a physical count.

We report an unannounced manual headcount audit run against the sensor network's own occupancy index across nine weeks and 15 of the University's 21 registered corrals, treating simultaneous physical counts as ground truth for what the index is supposed to be estimating.

Our contributions, each stated with the hedging a single-term audit deserves:

- We estimate the correlation between the sensor’s rolling occupancy index and an independently observed manual headcount across 312 paired samples, and find it consistent with the index tracking something other than headcount in the corrals we audited.
- We show the shortfall tracks each corral’s average scooter dwell time rather than sampling noise, with slow-turnover corrals over-represented and high-turnover corrals under-represented at the top of the index’s own capacity-allocation ranking.
- We report an ablation removing the index’s internal smoothing window, and find the dwell-time pattern persists under instantaneous sampling, suggesting it is a property of what the sensor observes rather than of any one calibration parameter.

Related work

Occupancy sensing for parking and kerb-side infrastructure is a mature engineering literature in its own right, spanning magnetometer arrays validated against wireless-signal cross-checks [1], video-based detection pipelines [2], and drone-mounted vision models trialled against aerial imagery of car parks [3]. That literature treats detection accuracy — did the sensor register a vehicle correctly — as the object of study more often than it asks what an accurate per-sample detection adds up to once aggregated into a planning-facing index, a question closer to the spatio-temporal demand-modelling tradition established for curbside parking [4] and extended to compositional occupancy-count data including bike-share stations [5], alongside broader efforts to strengthen the sensing layer bike-share systems already carry [6].

Shared micromobility parking has its own applied literature, largely concerned with compliance rather than sensing: denser corral networks measurably raise parking compliance [7], sidewalk congestion and built-environment factors measurably lower it [8], and the introduction of mandatory corrals at one university campus measurably lowered ridership even as compliance rose [9]. Design pilots [10] and cross-city implementation guidance [11] both treat corral capacity as a planning lever without examining what instrument, if any, sets it; recent demand-

prediction [12] and rebalancing work, including autonomous-vehicle-assisted [13] and LLM-augmented [14] approaches, take a network’s sensor feed as a trusted input rather than an object of validation, and equity-focused rebalancing work raises a related question about who a mis-set allocation costs [15]. Our audit treats the feed itself as the thing under test, following work asking what a corral network’s compliance figures are actually built from [16]. The broader question of what a monitoring instrument comes to measure once an institution starts allocating resources against it draws on an older account of audit regimes reshaping the practices they observe [17] and on formal treatment of how competitively allocated proxies degrade under the pressure of being used that way [18].

Method

The University’s corral occupancy sensor is an infrared beam-break array mounted along the kerb rail of each corral, one emitter-receiver pair per marked bay. Each pair samples presence every 90 seconds; Facilities’ capacity-planning dashboard displays a smoothed occupancy index computed as an exponentially weighted moving average of the per-sample presence fraction, scaled to the corral’s bay count:

$$I_t = \alpha p_t + (1 - \alpha) I_{t-1}$$

where p_t is the fraction of a corral’s bays showing presence at sample t and α is set, per the vendor’s default configuration, to give the smoothing window an effective half-life of roughly fifteen minutes. Facilities’ term-end capacity review ranks corrals by mean I_t over the preceding term and directs the following term’s bay expansion to the corrals ranking highest.

We selected 15 of the University’s 21 registered corrals for audit, excluding six for which field-worker access or sensor uptime was unreliable across the audit window. Over nine weeks, two research assistants conducted unannounced walkthroughs at randomised times across each corral’s operating hours (07:00–22:00), physically counting scooters present at the instant of arrival and recording the sensor’s own I_t reading, read directly from the corral’s kerbside display panel, for the same minute. Walkthrough timing was drawn independently per corral per week from a uniform distribution over operating hours, subject only to a minimum ninety-minute gap between successive visits to the same

corral, yielding 312 paired observations after excluding 18 visits where the display panel was obscured or faulted.

For each corral we additionally computed mean scooter dwell time from the sensor's own event log (arrival and departure timestamps per bay), giving an independent per-corral measure of turnover unavailable to the audit team in the field. Agreement between I_t and the manual headcount was estimated with a fitted linear model and Spearman's rank correlation across all 312 paired observations, and separately, corral by corral, against each corral's own mean dwell time, to test whether the shortfall between sensor and headcount tracked turnover specifically.

The ablation reported in Results recomputes I_t with $\alpha = 1$ — an unsmoothed instantaneous read, discarding the exponential window entirely — against the same 312 audit timestamps, isolating whether the smoothing window itself, rather than what each sample measures, drives the shortfall from headcount.

Results

Across the 312 paired observations, the sensor's occupancy index and the manual headcount correlate only moderately (Spearman's $\rho = 0.49$; a fitted linear model explains 22% of headcount variance, Figure 1). The relationship is not simply noisy: when we regress the sensor-headcount residual against each corral's own mean scooter dwell time, computed independently from the sensor's event log, dwell time explains most of what the paired-observation model leaves unexplained. Corrals in the top tercile of mean dwell time read, on average, 1.6 index points higher than their headcount alone would predict; corrals in the bottom tercile read 1.1 points lower.

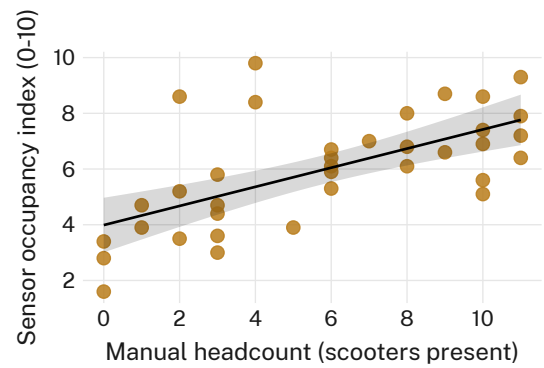


Figure 1: Manual headcount against the sensor's occupancy index across 312 paired observations: a fitted linear model explains 22% of headcount variance.

The consequence for capacity planning is direct. Facilities' term-end review ranks corrals by mean index and directs bay expansion to the top five. Of the five corrals the index ranked highest this term, only two also appear in the top five by mean manual headcount; the two corrals with the longest mean dwell time in the audited sample ranked first and second by index despite ranking ninth and thirteenth by headcount. Figure 2 illustrates the mechanism at one high-turnover corral audited across a full operating day: physical headcount rises and falls sharply around the two daily commuter peaks, while the smoothed index, averaging over scooters that individually never stay long enough to move the fifteen-minute window far, stays comparatively flat through both.

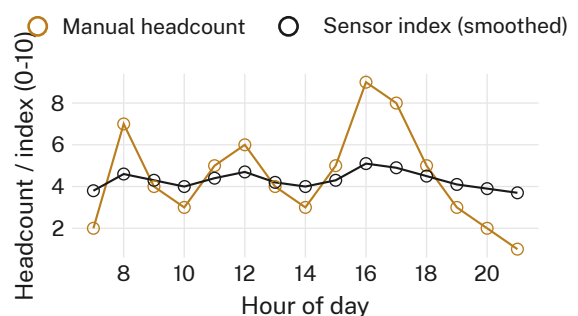


Figure 2: One high-turnover corral across a full operating day: manual headcount tracks the commuter peaks; the smoothed index barely moves.

We ran one ablation on the index itself: recomputing it with the smoothing window removed ($\alpha = 1$, an unsmoothed instantaneous read) against the same 312 timestamps. Median per-corral correlation with manual headcount was unchanged across the fifteen audited corrals,

0.35 smoothed and 0.35 unsmoothed, a difference well inside the spread within either condition (Figure 3). We retain the vendor’s default smoothing window in the discussion that follows; removing it does not materially change what the index tracks.

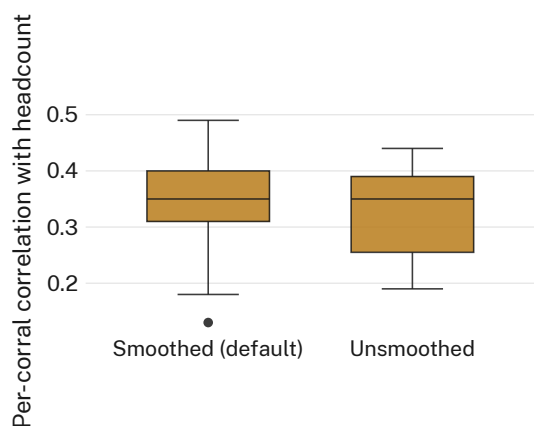


Figure 3: The ablation that changes nothing: per-coral correlation with manual headcount, smoothed and unsmoothed, holds at the same median (0.35 vs 0.35).

Two further observations round out the record without altering it. First, the six corrals excluded from the audit for unreliable sensor uptime were, when we later inspected Facilities’ own maintenance log, disproportionately drawn from the network’s lowest-dwell, highest-turnover locations — an availability pattern that, if it held across the full network, would bias any index-only capacity review toward the corrals already over-represented in our findings. Second, no audited corral’s dwell-time tercile changed across the nine-week window, suggesting turnover is closer to a stable property of a corral’s location than a response to any single week’s demand.

Discussion

The sensor’s occupancy index was never specified, as far as we can establish from Facilities’ own documentation, against a target of “number of scooters present”; the vendor’s material describes it only as an “occupancy indicator,” a term the index satisfies under a dwell-weighted reading as readily as under a headcount reading. The index may not be wrong about what it measures; the trouble is a capacity-allocation process built around its ranking behaving as though the two readings were interchangeable, and our audit suggests they are not, at least in the corrals and term we sampled.

We read the pattern as consistent with a familiar shape in institutional measurement: an instrument adequate for its narrower original purpose — flagging that a corral’s bays are in use at all — accrues a second, unexamined role once it becomes the only number the capacity review consults [17], and a competitive allocation built on a proxy tends to reward whatever the proxy rewards rather than whatever the allocation was meant to reward [18]. Nothing about the sensor needs to change to keep doing its original job; what would need to change is the assumption, currently load-bearing in the capacity review, that the index and the headcount measure the same thing.

Limitations

Our sample covers 15 of the University’s 21 registered corrals over a single nine-week window, and we make no claim the dwell-turnover pattern generalises beyond the term audited, though the pattern held consistently across all three dwell-time terciles in both the first and second halves of the window, which is more stability than nine weeks strictly warrants claiming. Fieldworkers were not announced to Facilities or to corral users, but were, for practical reasons, aware of the audit’s purpose, an asymmetry we consider unlikely to have altered scooter placement behaviour nobody involved knew was being observed. We audited the index’s correlation with headcount rather than redesigning the index itself; a headcount-weighted alternative remains untested.

Conclusion

An occupancy index built to flag whether a corral’s bays are in use, and a physical headcount of the scooters actually parked there, correlate only moderately across the corrals and term we audited ($\rho = 0.49$), with the short-fall tracking each corral’s scooter dwell time rather than sampling noise. Facilities’ current capacity-allocation ranking, which consults the index alone, sent only two of its top five corrals to locations that also ranked in the top five by manual headcount. An ablation removing the index’s smoothing window leaves the pattern unchanged, locating the shortfall in what the sensor observes rather than in how it averages. We suggest a headcount-weighted term audit, even an occasional one, accompany the index ahead of any future capacity-allocation round.

Future work might extend the audit to the full 21-corral network across a full year.

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